

Biomechanical Comparison of Patellar Fixation Techniques in Medial Patellofemoral Ligament Reconstruction

A Pediatric Cadaveric Study

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Background: The biomechanical properties of different medial patellofemoral ligament (MPFL) reconstruction (MPFLR) techniques have been described in adult cadaveric models. However, little is known about the biomechanical properties of these techniques in pediatric specimens.

Purpose: To (1) evaluate the biomechanical fixation strength of pediatric patellae in the context of MPFLR and (2) compare the performance of suture-anchor (SA) and bone-tunnel (BT) patellar fixation.

Study Design: Controlled laboratory study.

Methods: Ten pediatric cadaveric patellae from 6 donors (3 male and 3 female, mean 9.8 ± 0.75 years, range 9-11 years) were utilized. For the all-SA group, two 1.8-mm all-SA, each loaded with sutures, were placed on the medial side of the patella at the articular cartilage margin. For the BTs, two 2-mm tunnels were drilled from the medial side of the patella to the anterior patellar surface. No. 2 braided suture was then passed through the tunnels. Both constructs were loaded into a servomechanical testing apparatus, and a suture was tied to the top frame such that pullout would be in the same plane as the fixation construct, oriented to approximate the plane of the native MPFL insertion.

Results: BT constructs demonstrated a mean ultimate failure load of 169.8 ± 88.6 N, significantly higher than the 55.3 ± 20.1 N observed for SA constructs ($P = .002$). No significant difference in stiffness was found between BT (6 ± 1.9 N/mm) and SA (6.4 ± 1.7 N/mm; $P = .647$). The mechanisms of failure were anchor pullout for all SAs and suture cheese-wiring for all BTs.

Conclusion: In this pediatric cadaveric model, BT fixation demonstrated higher ultimate failure loads than the SA fixation, while construct stiffness was similar between techniques.

Clinical Relevance: To our knowledge, this is the first biomechanical study of MPFL patellar fixation reconstruction techniques performed in pediatric specimens. To reduce the risk of patellar graft fixation failure, the relatively low load-to-failure of all-SA fixation demonstrated in this study may be considered when treating pediatric patients with lower bone density. Moreover, biomechanically superior small-diameter oblique BTs with low fracture risk may be considered to maximize time-zero fixation strength compared with SAs in the pediatric population. However, these results are limited by evaluating only a single all-SA design and a testing methodology that focused on time-zero fixation strength, without cyclic loading or in vivo healing.

Keywords: bone tunnel testing; medial patellofemoral ligament reconstruction; patellofemoral instability; suture anchor fixation

Primary^{5,19} and recurrent⁸ patellar dislocations are common injuries in the pediatric population. The medial patellofemoral ligament (MPFL) serves as the primary soft

tissue mechanism restraining lateral translation of the patella, before it engages with the trochlear groove, from 0° to 30° of knee flexion.^{9,13,22,37,46} Reconstructive surgery of the MPFL has proven effective for many patients with patellar instability.^{3,17,18,23,27,28,30,43} While adult MPFL reconstruction (MPFLR) demonstrates an excellent pooled recurrent instability rate of only 1.2% (95% CI, 0.3-2.1),

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estimates in the skeletally immature population have greater variability, with pooled recurrence rates of approximately 2% but a wider confidence interval (95% CI, 0-9), reflecting uncertainty in pediatric outcomes.^{1,2,20,38,43,49,50} In pediatric MPFLR, surgeons face the added difficulty of physeal avoidance at the femoral attachment, which may influence outcomes compared with adult populations. However, in both populations, patellar fixation has been reported as a source of reconstruction failure.⁴⁴

Several MPFLR patellar graft fixation techniques have emerged,^{11,12,51} including BT fixation, cortical surface SA fixation, and other cortical surface approaches, such as periosteal suturing. Because robust clinical trials with long-term follow-up are lacking, patellar fixation techniques have instead been evaluated based on their biomechanical performance.^{23,23,27-33} BT fixation methods for MPFLR can use convergent, divergent, socket, or transverse BTs to accommodate both implant-based and implant-free grafts.^{6,14,27,29,31,42} While these techniques demonstrate superior fixation biomechanics to patellar cortical surface SA fixation techniques, they are associated with higher rates of postoperative patellar fractures,^{4,11,31,42} particularly with larger diameter tunnels and use of screw implants.^{15,29,33,42,47} Patellar cortical surface fixation techniques, which require partial tunnels with simple sutures or SAs, present a bone-sparing alternative with lower risk of fracture.^{12,21,39,41} However, given the trade-off between fracture risk and biomechanical strength, there is no clear consensus on the optimal patellar fixation technique.²⁴

A few biomechanical studies comparing patellar fixation techniques have been conducted in both animal models and adult cadaveric specimens.^{37,38,40-42} However, to our knowledge, no such studies have been conducted in skeletally immature specimens. Pediatric bones exhibit distinct characteristics from adult bones, including lower mineral density and ongoing growth,²⁷ which can significantly influence the interaction between surgical fixation devices and the bone. Although BT fixation has demonstrated superior biomechanical performance compared with SA fixation in adults, the more pliable, less dense nature of pediatric bone may alter this relationship, potentially compromising MPFLR stability.¹⁰ Without a direct comparison of SA and BT constructs in pediatric specimens, the advantages and drawbacks of each technique remain unclear. This discrepancy underscores the need for biomechanical studies tailored to pediatric populations to optimize surgical outcomes and address the unique challenges posed by the developing skeletal system.

This study aimed to compare, in a pediatric cadaveric model, the biomechanical fixation strength of 2 patellar MPFLR constructs: SA anchor fixation and BT. By focusing on pediatric tissue, this research seeks to address a critical gap in the current literature on MPFLR, paving the way for pediatric-specific surgical treatments that consider the unique properties of skeletally immature tissue. We hypothesize that BT fixation will show no statistically significant difference in load-to-failure or stiffness compared with SA fixation in a pediatric cadaveric model.

METHODS

Specimens

Institutional review board approval was not required, as this study was conducted on cadaveric tissue without patient identifiers, no genetic information was analyzed, and familial consent was obtained for research purposes at the time of tissue donation. Ten fresh-frozen cadaveric knees were obtained from 6 pediatric donors (3 males and 3 females, 9.8 ± 0.75 years). At the tissue bank laboratory (Allosource), the extensor mechanisms of each knee were dissected (Figure 1A), and the patellae were inspected. Any patellae with signs of knee trauma, malformation, or surgery were excluded. The number of specimens selected was based on tissue availability, which is severely limited for pediatric tissue. The selected samples were frozen and shipped back to a biomechanical laboratory for testing.

Testing Protocol

After 24 hours of thawing at room temperature, each patella was dissected from the extensor mechanism, and its dimensions were measured. A custom aluminum plate was manufactured to mount the patella to the servomechanic test frame (Instron Model 5944, Instron). Because of the size of the pediatric patellae, a standard clamp provided insufficient fixation. To prevent patellar movement during testing, the patella was fixed to the mounting plate with 2 bolts (a 6-mm bolt at the center of the patella and a 3-mm bolt in the distal third at the centerline). The bolt was positioned away from the medial fixation sites to minimize interference with anchor or tunnel behavior, and both fixation techniques were tested under identical mounting conditions.

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Ethical approval was not sought for the present study. This study was conducted on cadaveric tissue without patient identifiers, no genetic information was analyzed, and familial consent was obtained for research purposes at the time of tissue donation.

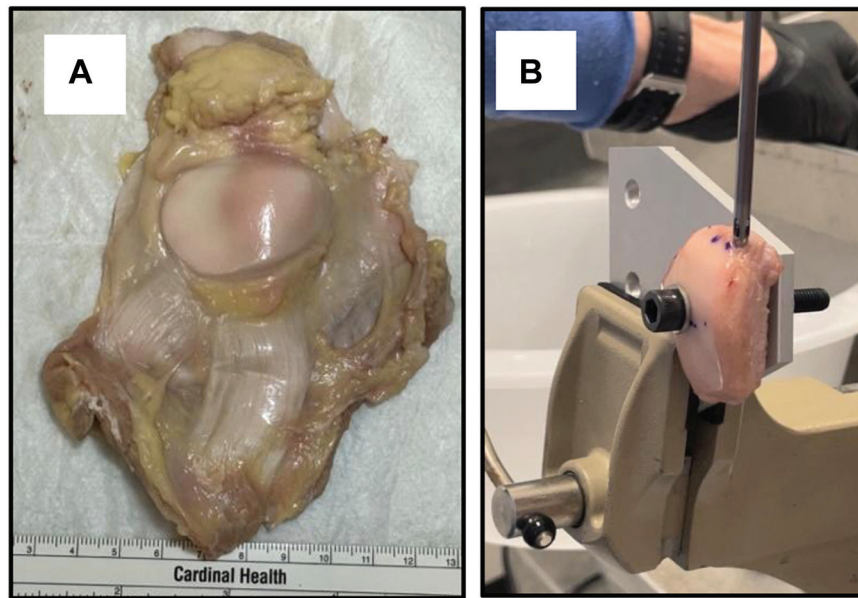


Figure 1. (A) Extensor mechanism with patella shown after thawing before final dissection. (B) Patella in the mounting plate in preparation for SA pilot hole drilling. SA, cortical surface suture anchor.

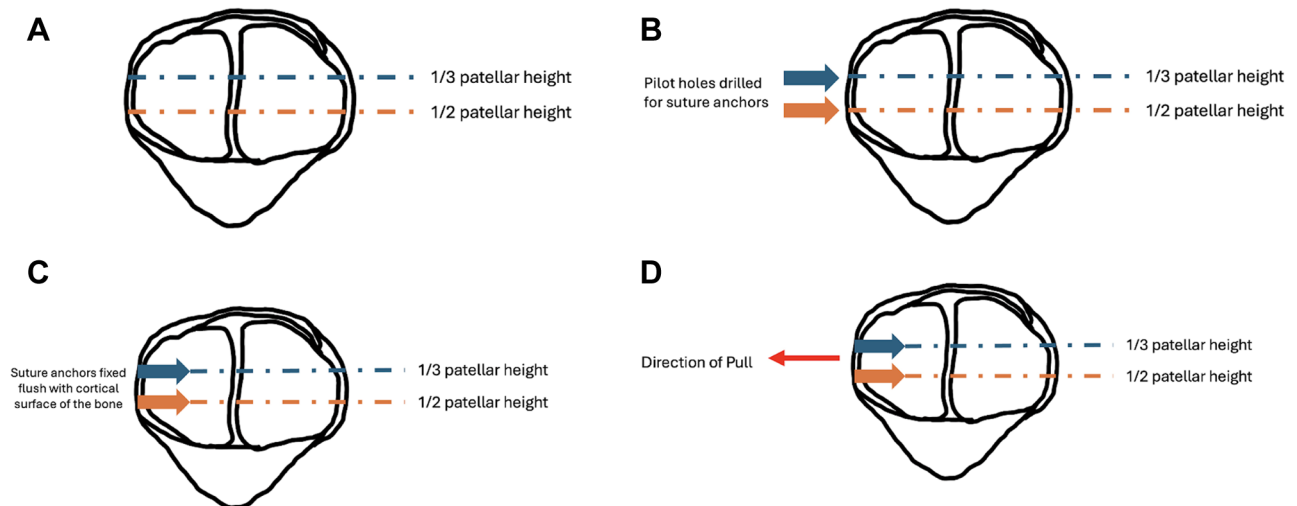


Figure 2. A stepwise protocol was followed to place two 1.8-mm suture anchors on the superomedial aspect of the patella. (A) Anchor placement sites were identified based on 1/3 and 1/2 of patellar height. (B) A 1.8-mm drill bit was passed through a drill guide at both anchor locations to create a pilot hole. (C) The suture anchors were fixed flush with the cortical surface of the bone. (D) Sutures were pulled medially, along the same plane as the suture anchors.

Once the patella was securely attached to the mounting plate, the plate was clamped into vice grips to provide stability for SA placement (Figure 1B).

Two 1.8-mm SAs (Q-Fix, Smith and Nephew) were placed on the superomedial patella at positions 1/3 and 1/2 of the total measured superior-to-inferior axis (Figure 2A). To place the anchors, a 1.8-mm drill bit was passed through the drill guide to create a pilot hole (Figure 2B). The SA device was passed through the drilled guide into

the pilot hole and deployed (Figure 2C). Light tension was applied to the SA sutures to verify fixation within the patella. Once both SAs were fixed, the mounting plate was clamped to the bottom of the test frame. The sutures of 1 SA were tied to a bar at the top of the testing frame in such an orientation that the direction of pull was in the same plane as SA insertion (Figures 2 and 3). Given that SA fixation can be performed with either 1 or 2 anchors, each anchor was tested individually to isolate the anchor-

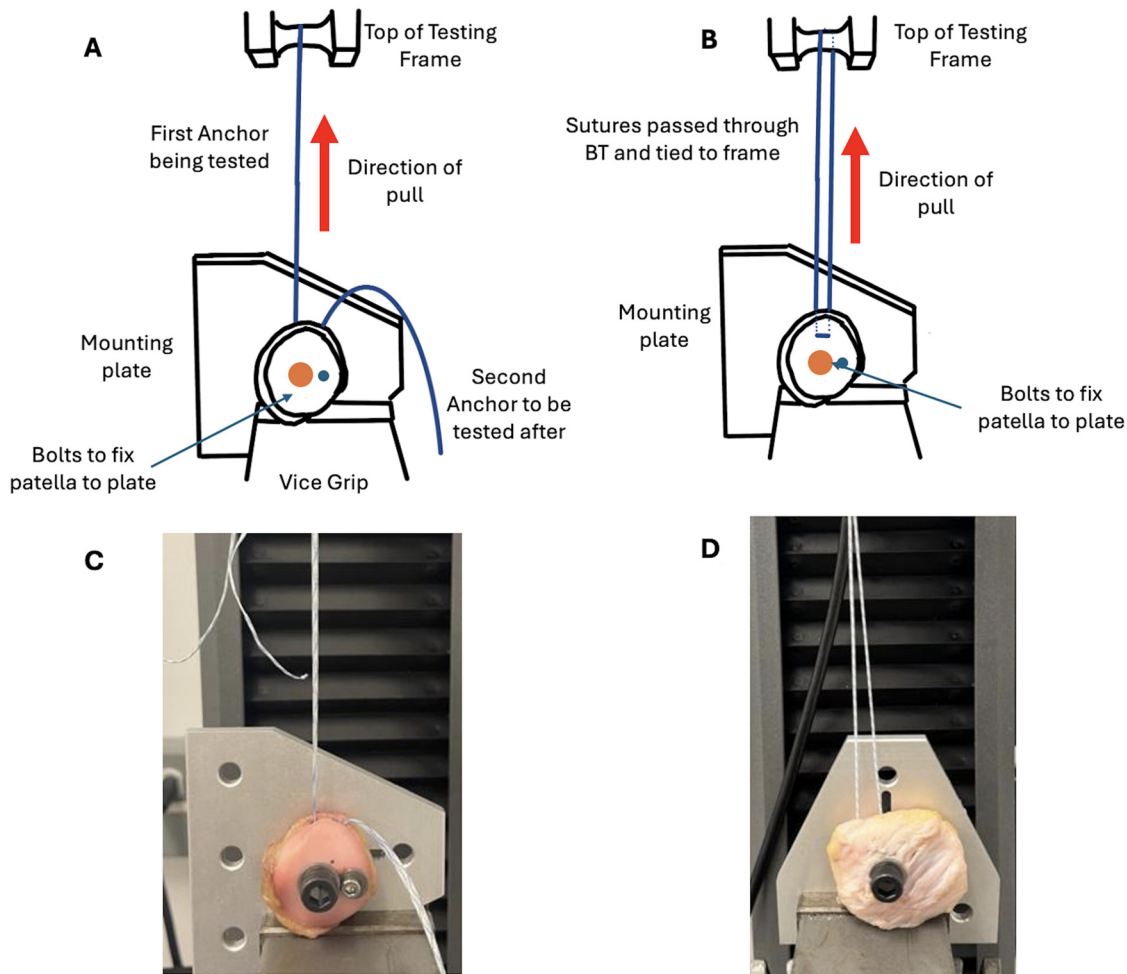


Figure 3. Testing protocol setup. (A) Diagram showing the loading protocol used to pull each suture anchor to failure. (B) Loading protocol to pull the sutures threaded through the BT. (C) Photo of patellar loading for suture anchors. (D) Photo of patellar loading for the bone tunnel. BT, bone tunnel.

bone interface and quantify independent pullout strength without confounding by load-sharing effects. This approach was intended to represent a conservative, minimum time-zero fixation strength.

Once loaded into the testing apparatus, all specimens underwent a 10-cycle preconditioning period from 5 to 25 N to reduce tissue hysteresis. Immediately after preconditioning, specimens were loaded at 10 mm/min until failure, which was defined as a sudden 10% drop in peak force. The load-to-failure phase was recorded on video to document the failure mechanism. Stiffness was calculated as the slope of the best-fit linear regression line applied to the linear portion of the load-displacement curve during monotonic loading, excluding the preconditioning cycles and postpeak failure region. The linear region was identified as the segment exhibiting the highest linearity (maximum R^2) before ultimate failure (Figure 4). The above steps were repeated for the second SA.

After testing both SAs, the patella was removed from the mounting plate to place the BTs. Two 2-mm BTs

were drilled from the medial border of the patella at the cartilage margin through the anterior patellar surface. The holes were placed at one-half the distance between the 2 SA holes and an equivalent distance distal to the distal-most SA (Figure 5, A and B). No. 2 braided suture (Ultrabraid No. 2, Smith & Nephew) was passed through the tunnels and tied identically as the SAs, with the patella reattached to the mounting plate and test frame (Figures 3 and 5). The servomechanical preconditioning and load-to-failure protocol was the same for the BTs as for the SAs.

Statistical Analysis

Data were collected and coded in Microsoft Excel, Version 16.81 (Microsoft Corp). All statistical analyses were performed using SPSS, Version 28.0.1.0 (IBM Corp). Normality of data distribution was assessed with the Shapiro-Wilk test. Continuous variables did not depart significantly from normality ($P > .05$) and are presented as means

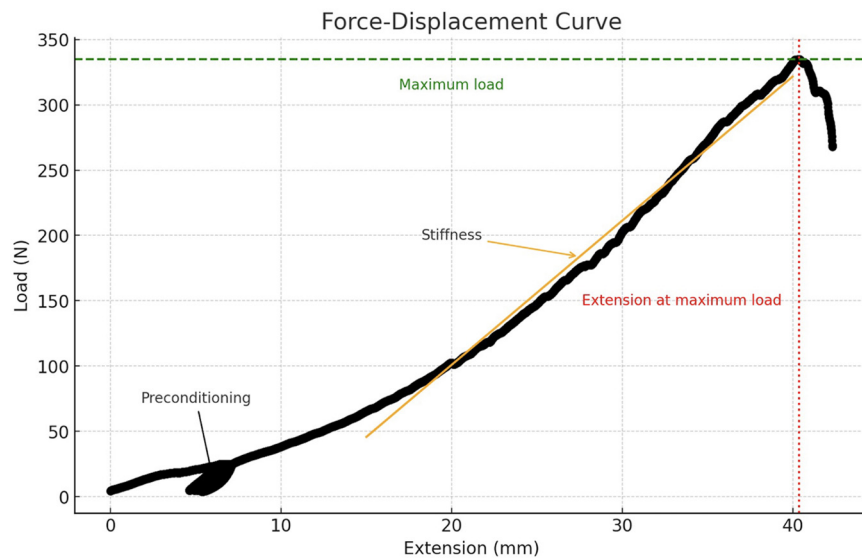


Figure 4. Force displacement curve. Example of the data from specimen 6 (BT) captured throughout the testing protocol. The maximum load and stiffness shown reflect this individual specimen and are not representative of the cohort mean values. Preconditioning is first performed between 5 and 25 N of force, then the specimen is pulled to failure at a rate of 10 mm/min. Construct stiffness (N/mm) is defined as the maximum slope of the load displacement curve. Maximum load is defined as the maximum force experienced before rupture of the test specimen. The stiffness line shown is provided for illustrative purposes, demonstrating the region used for stiffness calculation. BT, bone tunnel.

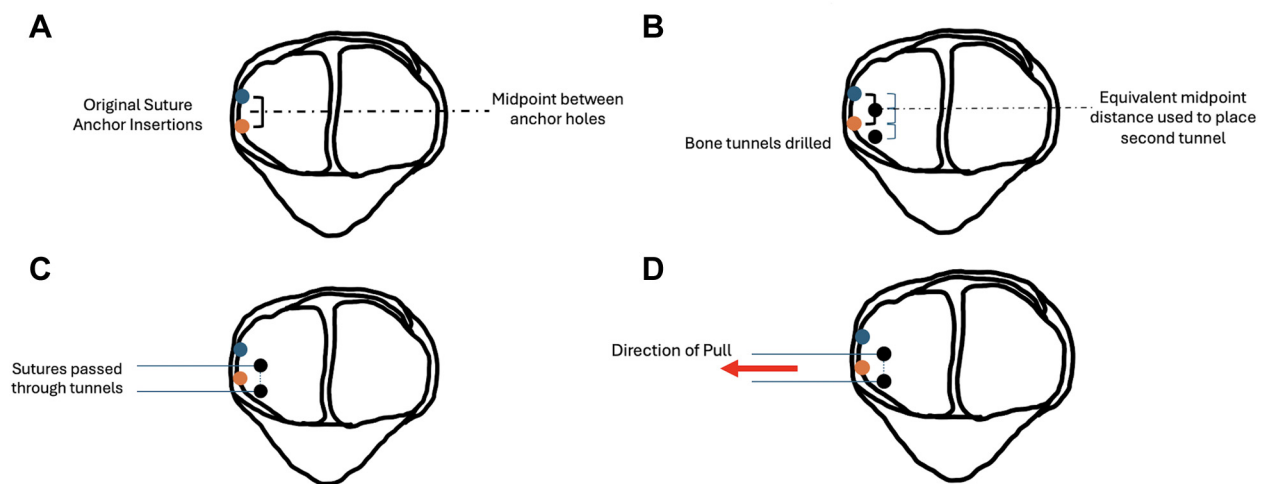


Figure 5. Bone tunnel placement and testing protocol. (A) The first bone tunnel was drilled from the medial patellar border to the anterior surface at the midpoint between the 2 previously placed suture anchors, establishing its superior-inferior position. (B) The second bone tunnel was placed the same distance distal to the most inferior suture anchor. (C) No. 2 braided sutures were passed through both tunnels and secured to the testing apparatus. (D) Sutures were pulled medially in the same plane as during suture anchor testing to maintain consistency in loading direction.

and standard deviations. Descriptive statistics were calculated. Paired *t* tests were used to compare the mean load to failure and stiffness between the SA and BT techniques. Statistical significance was set at .05. Also, 95% CIs for the mean paired difference between BT and SA fixation were calculated for load-to-failure and stiffness to characterize the uncertainty in effect-size estimates.

RESULTS

Proximal (58.22 ± 20.90 N) and distal (52.37 ± 22.42 N) SA loads to failure did not differ significantly ($P = .554$); thus, they were averaged together for comparison with BT. The mean load to failure for BT fixation was 169.81 ± 88.58 N, which was significantly higher than the $55.29 \pm$

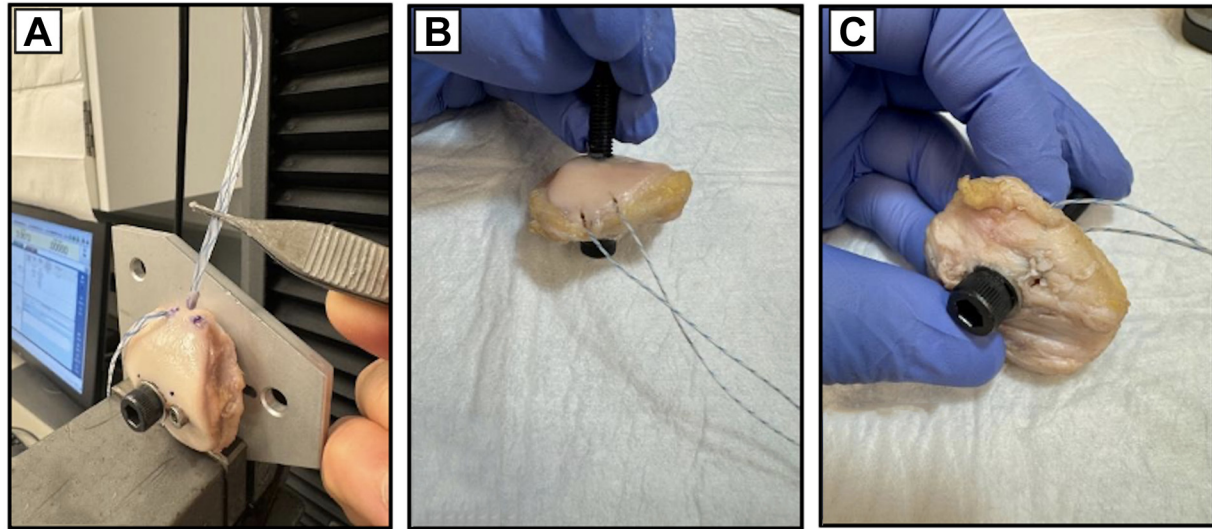


Figure 6. (A) Anchor pullout mechanism of failure for suture anchors. (B) Cheese-wiring mechanism of failure for BT, posterior aspect. (C) Cheese-wiring mechanism of failure for BT, anterior aspect. BT, bone tunnel.

20.11 N observed for averaged SA fixation (mean difference [MD], 114.51 N [95% CI, 54.33 to 174.69 N]; $P = .002$). Stiffness was similar between BT (6.01 ± 1.93 N/mm) and SA (6.37 ± 1.73 N/mm), with no significant difference detected (MD, -0.36 N/mm [95% CI, -2.07 to 1.36 N/mm]; $P = .647$).

The mechanisms of failure were anchor pullout (Figure 6A) for all 20 SAs and suture cheese-wiring (Figure 6, B and C) for all 10 BT constructs. There was no evidence of patellar fracture or tunnel collapse in either group. Specimen demographic, dimensional, and biomechanical testing data are shown in Table 1.

DISCUSSION

The main finding of this study was that, in pediatric patellae, the BT fixation technique achieved higher ultimate failure loads than the SA fixation technique, with statistically similar stiffness. To our knowledge, no MPFLR patellar fixation biomechanical studies have been performed in human pediatric specimens. By focusing on pediatric tissues, this study addresses a critical gap in the current literature regarding skeletal immature patients, a population with the highest rates of primary and secondary patellar instability,^{19,25,35,40} and provides insights that may influence surgical techniques for this group.

In the existing literature, several studies have assessed the biomechanics of different fixation constructs in adult cadaveric patellae. As previously reported by Raoulis et al,³⁶ adult BT constructs failed at 263.2 ± 20.1 N and SA constructs at 243 ± 41.9 N; in our pediatric cohort, the corresponding values were 169.8 ± 88.6 N and 55.3 ± 20.1 N, respectively, reflecting the diminished ossification and altered material properties of skeletally immature bone. Similarly, Raoulis et al³⁶ reported BT and SA stiffnesses of 21.4 ± 9.5 N/mm and 18.7 ± 3.4 N/mm, nearly

3 times greater than our pediatric measurements of 6 ± 1.9 N/mm and 6.4 ± 1.7 N/mm. Anchor pullout remained the primary failure mode for SA constructs in pediatric patellae, mirroring findings from adult studies and indicating that, although bone maturity reduces pullout strength, it does not alter the failure mechanism. By contrast, Raoulis et al³⁶ reported 1 case of BT collapse and predominantly graft-suture interface failures in adult BT constructs. In our pediatric specimens, however, BT constructs always failed due to a cheese-wiring effect, in which the suture cuts through the softer, immature bone. This distinction underscores the unique biomechanical properties of pediatric bone. It reinforces the need to evaluate multiple patellar fixation techniques, since construct performance and failure modes may differ in immature specimens.

Compared with 12-month-old porcine patellae, pediatric specimens exhibited markedly lower biomechanical properties. Zhao and Zang⁵² reported ultimate failure loads of 496.2 ± 93.2 N for BT constructs and 399.4 ± 105.4 N for SA constructs in porcine samples, with stiffness values of 55.4 ± 7.9 N/mm and 72.1 ± 10.6 N/mm, respectively. These loads and stiffnesses exceed our pediatric measurements by more than 3-fold. Although porcine models are widely used to approximate the pediatric knee,^{16,26} our study, the first to evaluate MPFL patellar fixation in human pediatric cadaveric tissue, demonstrates that this model substantially overestimates fixation strength and stiffness, limiting their validity for preclinical assessment.

Although the adult cadaveric study by Raoulis et al³⁶ similarly reports higher mean failure loads for BT versus SA constructs, the difference did not reach statistical significance. However, in our pediatric specimens, BT fixation achieved a significantly greater ultimate failure load than SA fixation, while construct stiffness remained comparable. Raoulis et al³⁶ reported mean failure loads of 263.2 ± 9.1 N for BT and 243 ± 41.9 N for SA in adult specimens,

TABLE 1
Specimen Demographics, Patellar Measurements, Load to Failure, and Stiffness^a

SIN	Age	Sex	Side	Patellar Width, mm	Patellar Height, mm	Patellar Area, mm ²	Technique	Load to Failure, N	Stiffness, N/mm
1	9	M	L	31.33	28.27	442.85	SA	89.87	8.92
							BT	85.56	3.71
			R	33.26	28.01	465.81	SA	53.12	6.81
							BT	97.14	3.85
2	9	M	L	41.86	30.07	629.37	SA	58.85	6.06
							BT	201.86	6.74
			R	41.45	30.27	627.35	SA	75.21	5.48
							BT	273.62	8.65
3	10	F	L	37.82	30.13	569.76	SA	47.26	5.95
							BT	139.82	4.88
			R	37.82	30.13	569.76	SA	42.19	8.37
							BT	233.42	7.03
4	11	F	L	31.44	22.19	348.83	SA	25.83	3.96
							BT	85.19	5.47
			R	30.27	21.63	327.37	SA	28.82	4.53
							BT	85.99	4.47
5	10	M	L	48.73	37.57	915.39	SA	65.43	5.14
							BT	160.68	5.94
6	10	F	R	38.85	31.89	619.46	SA	66.38	8.47
							BT	334.8	9.36
Mean ± SD				37.28 ± 5.84	29.02 ± 5.84	551.59 ± 5.84	SA	55.29 ± 20.11	6.37 ± 1.73
							BT	169.81 ± 88.58	6.01 ± 1.93
P								.002	.647

^aResults obtained for BT versus SA fixation. The bold *P* value indicates statistical significance. Area calculated as a triangle ($0.5 \times \text{patellar width} \times \text{patellar height}$). BT, bone bridge; F, female; L, left; M, male; R, right; SA, suture anchor; SID, specimen identification number.

with no statistically significant difference between techniques. Other adult studies have described lower SA failure loads of 201.5 ± 63.1 N (Russ et al³⁹) and 229.5 ± 53.1 N (Saper et al⁴¹). In our pediatric cohort, BT constructs failed at 169.8 ± 88.6 N versus 55.3 ± 20.1 N for SA constructs ($P < .05$). Both this pediatric study and other adult investigations demonstrated no significant differences in stiffness between fixation methods. These results suggest that BT fixation provides superior time-zero load-bearing capacity in immature patellae and may be considered when higher initial fixation strength is required.

In porcine patellar models, BT fixation similarly produced significantly greater ultimate failure loads than SA fixation, despite the absolute strengths in these animal specimens being markedly higher than those we observed in pediatric knees. Lenschow et al²⁶ reported mean failure loads of 539.5 ± 159.6 N for BT constructs versus 401.5 ± 96.1 N for SA constructs ($P < .05$). Similarly, Zhao and Zang⁵² found that BT constructs failed at 496.2 ± 93.2 N compared with 399.4 ± 105.4 N for SA constructs ($P < .05$). These findings reinforce the superior load-bearing capacity of BT fixation across models, while emphasizing the need for caution when extrapolating porcine data to the pediatric human patella.

The superior fixation strength of BT constructs in pediatric patellae may confer added protection to the reconstructed MPFL during early rehabilitation.⁵² Clinical evidence supports this advantage: in a series of 29

pediatric patients treated with BT fixation, Zhang et al. reported no episodes of patellar dislocation. They demonstrated significant improvements in all clinical outcome scores from pre- to postoperatively.⁵¹ However, transpatellar drilling carries inherent risks. Malpositioned tunnels or excessively large diameters can act as stress risers and predispose patients to patellar fracture.^{32-34,45,48} However, Zhang et al⁵¹ assert that by minimizing tunnel diameter, the risk of patellar fracture remains low and is more than offset by the biomechanical advantages of BT fixation. Moreover, BT techniques may offer notable cost savings: while 2 SAs for MPFLR often exceed \$1000, a BT construct requires only standard No. 2 nonabsorbable sutures⁵² (typically around \$100). These economic considerations may further inform surgical decision-making in the pediatric population.

Despite lower fixation strength, SA constructs offer distinct advantages. Implant-based techniques are generally faster and technically less demanding than BT methods.⁴⁵ By avoiding transpatellar drilling, SAs may also reduce the risk of iatrogenic patellar fracture, although prospective clinical studies are needed to confirm this protective effect.⁴¹ However, there are also negative effects associated with SA placement in MPFL reconstruction. Some series have reported persistent postoperative pain and soft tissue irritation at anchor sites.⁴⁵ Furthermore, implant-related complications such as malposition or loosening of anchors are a concern in MPFLR.⁴⁵ In all

MPFLRs, approximately 1.1% require revision due to issues with surgical implants.⁴⁵

There are several limitations to this study. First, the SA and BT constructs were performed on the same patellae, not isolated to separate groups of patellae. This was done to preserve very limited and hard-to-obtain pediatric tissue. It is possible that the SA pilot holes and stresses of the SA biomechanical testing weakened the patellae before the BT construct was placed and tested. However, if this were the case, the ultimate failure loads and stiffness values measured during the BT phase of the study would have been even higher than those measured in this study. Therefore, the study's rejection of the null hypothesis would remain true. Second, this study was limited by a small sample size. Again, the availability of healthy, fresh-frozen, human cadaveric tissue from pediatric populations is extremely limited; the opportunity to expand the study's sample size would have been ideal, but was not possible given the limited donor pool. This limitation was mitigated by placing 2 SAs on each patella and reusing the same patellae for BT testing. The 95% CI for the MD in stiffness (-0.36 N/mm [95% CI, -2.07 to 1.36 N/mm]) spans zero and is narrow, suggesting that any true difference in stiffness between constructs is small, although uncertainty remains given the sample size. Third, the study's testing protocol (constant rate extension in a straight line parallel to SA or BT placement) does not exactly replicate physiologic knee movements. Both SAs and BTs might actually fail at higher loads under stresses exerted with normal human knee movements, since these forces are not always directed in line with the placement of the anchors or sutures. However, the testing protocol represents a worst-case scenario, reduces variation between samples, and limits confounders within this laboratory-based study. Moreover, this testing protocol did not incorporate cyclic loading and instead assessed time-zero fixation strength. As a result, the findings reflect initial construct strength rather than fixation behavior under repetitive or physiologic loading conditions. Fourth, unlike in vivo fixations that benefit from added strength because of healing, the constructs tested in this study were evaluated immediately after placement. Again, this represents a worst-case scenario and is more conservative than in vivo conditions. Fifth, this study only tested SAs and BTs themselves, rather than first fixing them to a tissue graft. The SA sutures were tied directly to the test frame and pulled; the BT construct used sutures rather than tissue grafts, which were passed through the tunnels. This limitation was a calculated decision based on the limited pediatric tissue available. However, the focus of this study was on the strength and properties of the pediatric patellae in the context of SA and BT constructs. Our study design allowed us to efficiently test the construct strength independently of MPFLR graft strength. However, this limits physiologic simulation, as graft incorporation may influence load distribution and failure mechanisms observed in vivo. Sixth, the mean age of specimens in this study (9.8 years) is younger than that of MPFLR patients.⁷ This could contribute to lower fixation strength values for both constructs. However, because both fixation

techniques were tested within the same specimens, the comparative relationship between constructs remains internally controlled. Lastly, this study evaluated only a single all-SA design. Although fixation biomechanics may vary across models and manufacturers, evaluating additional designs was not feasible because of the limited availability of pediatric cadaveric tissue.

CONCLUSION

To our knowledge, this investigation is the first to report the biomechanical performance of MPFL patellar fixation constructs in human pediatric specimens. The significant differences in ultimate failure load and comparable stiffness between BT and SA techniques provide quantitative benchmarks to guide fixation choice in pediatric MPFLR.

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